

SUPERSTORE SALES OVERVIEW & GEOGRAPHIC INSIGHTS

End-to-End Cloud Analytics Reference Document

AWS | Amazon S3 | AWS IAM | AWS Glue Crawler | AWS Glue Data Catalog
Amazon Athena | SQL Queries | Athena ODBC | Power BI | Power Query | DAX
Measures and Calculated Columns

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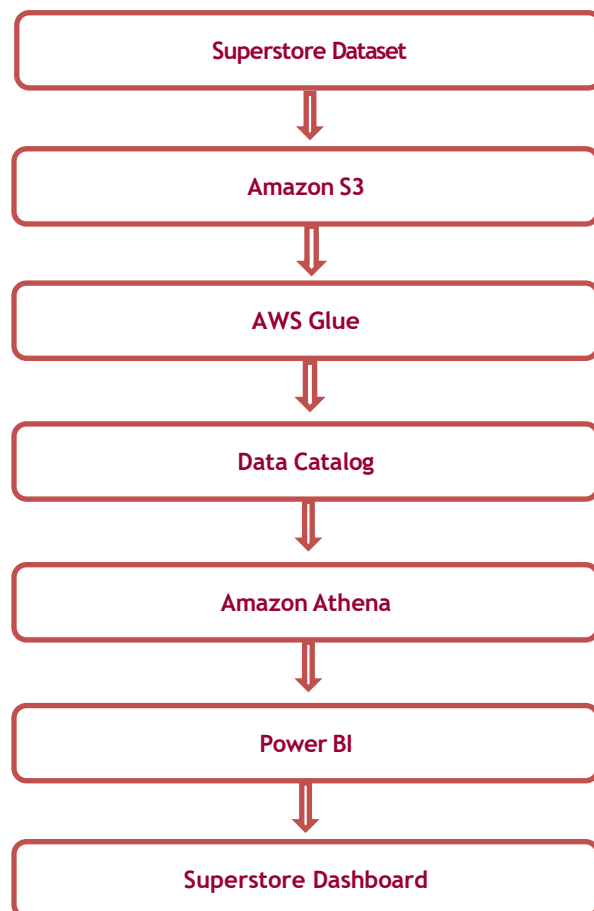
Batch Code: ANP-D6641

Course Name: DBA – Data and Business Analysis

PROJECT OBJECTIVE

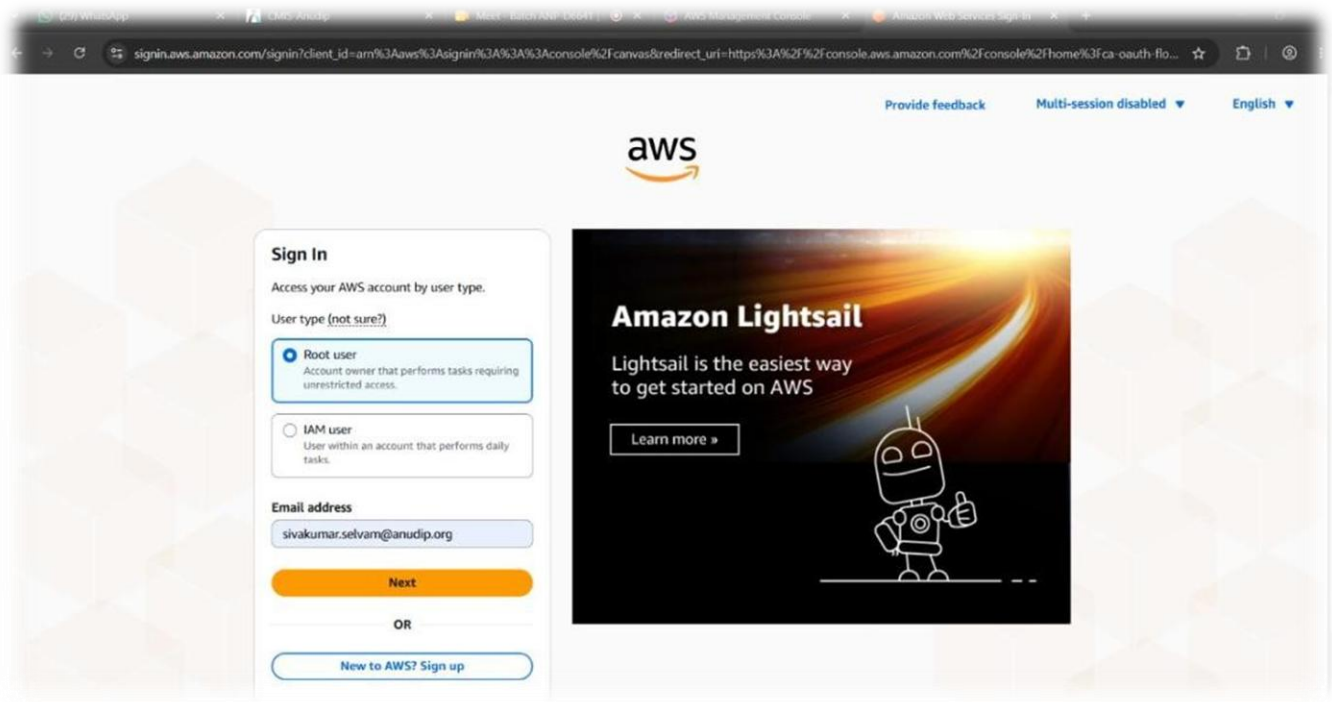
- The main objective of this project is to develop an end-to-end cloud-based data analytics solution using the Superstore dataset. The project involves storing and managing the raw dataset in Amazon S3, configuring AWS IAM to provide secure and controlled access to the required AWS resources, and using AWS Glue Crawler to automatically identify the data schema, create a data catalog, and organize the dataset for further analysis.
- The data is then connected to the analytics environment and processed using Power Query, where required data cleaning, transformation, formatting, and preparation activities are performed to improve data quality and consistency.
- After preparing the data, it is integrated with Power BI to perform data analysis and create meaningful visualizations. An interactive Power BI dashboard is developed in two pages: the first page provides a Sales Overview with KPI cards, charts, and key sales and profit metrics, while the second page presents Geographic and Customer Insights using location-based visuals, customer analysis, charts, and informative elements. The overall objective is to transform raw Superstore data into structured, interactive, and meaningful business insights that can support effective analysis and data-driven decision-making.

COMPLETE PROJECT WORKFLOW:



STEP 1: CREATE AWS MANAGEMENT CONSOLE ACCOUNT

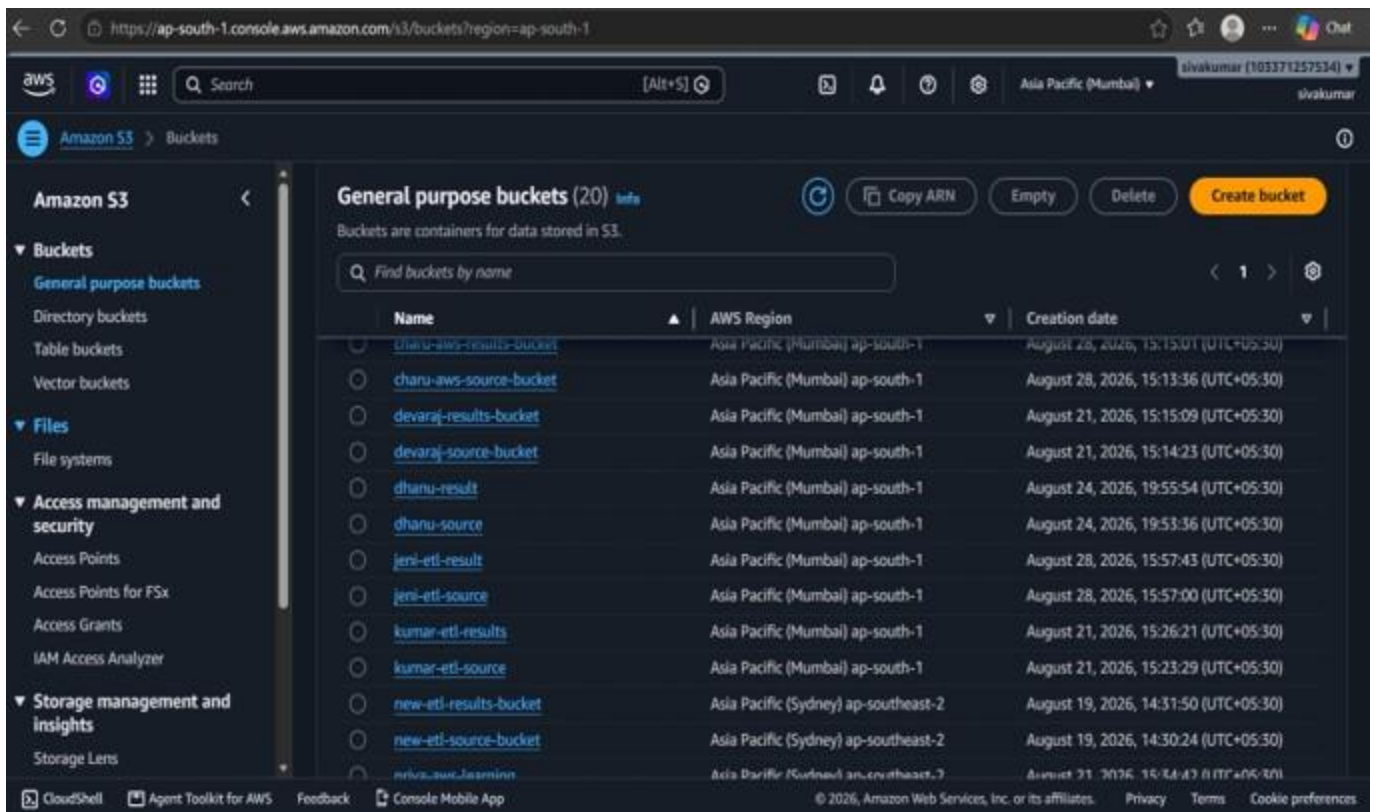
- Signed in to AWS Management Console using the account credentials.
- Enabled Multi-Factor Authentication (MFA) for account security.
- Verified successful sign-in and accessed the Console Dashboard.



STEP 2: CREATE S3 BUCKETS

- In the AWS search bar, searched for "S3" and opened the S3 service.
- Created two separate buckets:
 - **charu-etl-source-bucket**— to hold the raw dataset
 - **charu-etl-results-bucket** — to hold Athena query results
- Downloaded the sample Superstore dataset in .csv (comma-separated values) format.
- Uploaded the dataset inside the source bucket by creating a folder and adding the file.

S3 Location	Purpose
Source Bucket / Flight Data Folder	Stores the uploaded flight dataset
Query Result Location	Stores Athena execution results, when configured



STEP 3: CONFIGURE IAM ROLE

- Navigated to IAM -> Roles -> Create Role.
- Assigned the required permissions so the Glue service can access the S3 bucket.
- Selected the following permissions:
 - **AmazonS3FullAccess** — for bucket access
 - **AWSGlueServiceRole** — for Glue service access
 - (Alternatively, AdministratorAccess can be used for full permissions during learning/practice)
- Created and saved the IAM role for use by the Glue Crawler.

Step 2: Add permissions

[Edit](#)

Permissions policy summary

Policy name ↗	Type	Attached as
AmazonS3FullAccess	AWS managed	Permissions policy
AWSGlueServiceRole	AWS managed	Permissions policy

Step 3: Add tags

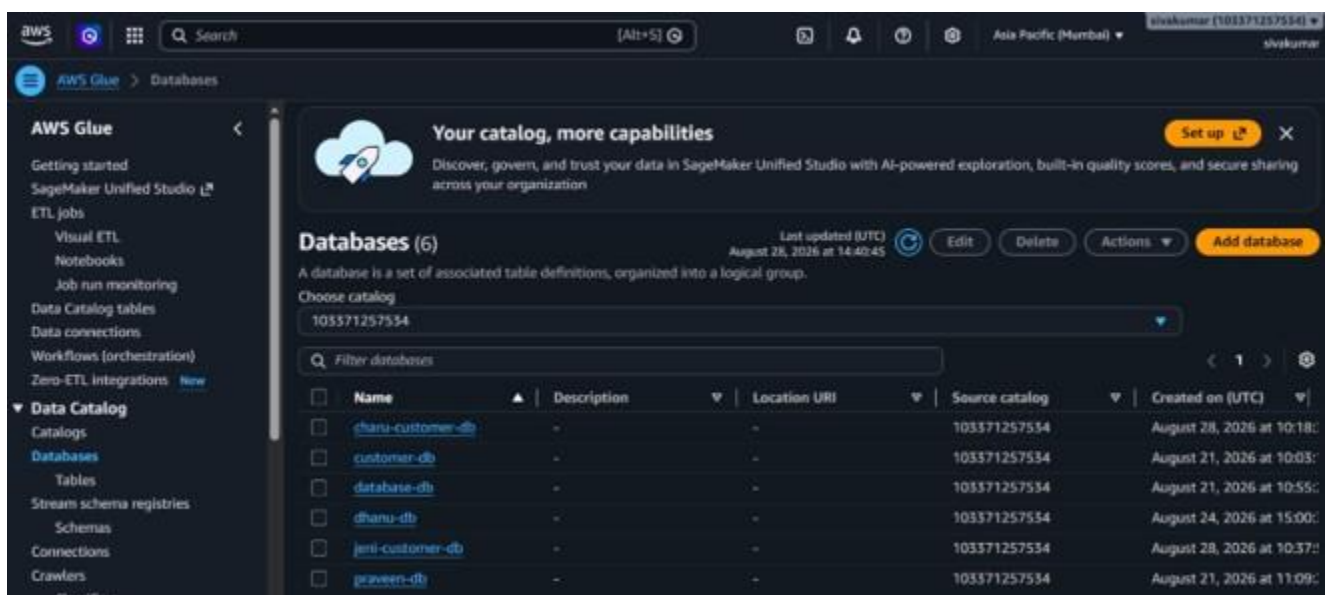
STEP 4: AWS GLUE – DATABASE, CRAWLER, AND TABLE

4.1 CREATE DATABASE

- In AWS Glue -> Data Catalog -> Databases, created a new database (e.g., superstore-db) to hold the catalog metadata.



- Opened AWS Glue and selected Data Catalog → Databases.
- Created a new database for the flight analytics dataset.
- Saved the database and verified that it appeared in the Data Catalog.



4.2 CREATE CRAWLER

- Navigated to AWS Glue -> Crawlers -> Add crawler.
- Entered crawler name: charu-etl-crawler

The screenshot shows the AWS Glue console's 'Add crawler' wizard. The left sidebar lists various AWS Glue services, with 'Crawlers' selected. The main panel displays 'Step 1: Set crawler properties'. A progress bar on the left indicates the steps: Step 1 (Set crawler properties), Step 2 (Choose data sources and classifiers), Step 3 (Configure security settings), Step 4 (Set output and scheduling), and Step 5 (Review and create). The 'Set crawler properties' section includes a 'Crawler details' box with a 'Name' field containing 'new-etl-crawler' and a 'Description - optional' field. Below this is a 'Tags - optional' section. At the bottom right are 'Cancel' and 'Next' buttons.

- On the "Choose data sources and classifiers" screen, selected Not yet for "Is your data already mapped to Glue tables?" since the schema needed to be auto-detected.
- Added the S3 data source:
 - Data source: S3
 - Location of S3 data: In this account
 - S3 path: s3://new-etl-source-bucket/customer-data/
 - Subsequent crawler runs: Crawl all sub-folders

The screenshot shows the 'Add an S3 data source' dialog box. At the top, there's a dropdown menu set to 'S3'. Below it is a 'Network connection - optional' section with a text input field and 'Clear selection' and 'Add new connection' buttons. The 'Location of S3 data' section has two radio buttons: 'In this account' (selected) and 'In a different account'. The 'S3 path' section has a text input field containing 's3://new-etl-source-bucket/customer-' and a 'Browse S3' button. Below the path field is a note: 'All folders and files contained in the S3 path are crawled. For example, type s3://MyBucket/MyFolder/ to crawl all objects in MyFolder within MyBucket.' The 'Subsequent crawler runs' section has three radio buttons: 'Crawl all sub-folders' (selected), 'Crawl new sub-folders only', and 'Crawl based on events'. There are also two checkboxes: 'Sample only a subset of files' and 'Exclude files matching pattern'. At the bottom are 'Cancel' and 'Add an S3 data source' buttons.

Choose data sources and classifiers

Data source configuration

Is your data already mapped to Glue tables?

☒ Not yet
Select one or more data sources to be crawled.

☐ Yes
Select existing tables from your Glue Data Catalog.

Data sources (1) [Q help](#)

The list of data sources to be scanned by the crawler.

Type	Data source	Parameters
S3	s3://new-etl-source-bucket/custom...	Recrawl all

[Edit](#) [Remove](#) [Add a data source](#)

Custom classifiers - optional

A classifier checks whether a given file is in a format the crawler can handle. If it is, the classifier creates a schema in the form of a StructType object that matches that data format.

[Cancel](#) [Previous](#) [Next](#)

- Selected the IAM role created in Step 3 (charu-etl-role) under "Configure security settings" so the crawler has permission to read S3 and write to the Glue Data Catalog.

Configure security settings

IAM role [Q help](#)

IAM permissions

To access other AWS services and resources you need an IAM role. Choose a role with the right permissions.

☐ Create default role ☒ Use another role

new-etl-role [Create new role](#)

[Edit role](#)

[View role details in IAM](#)

Lake Formation configuration - optional

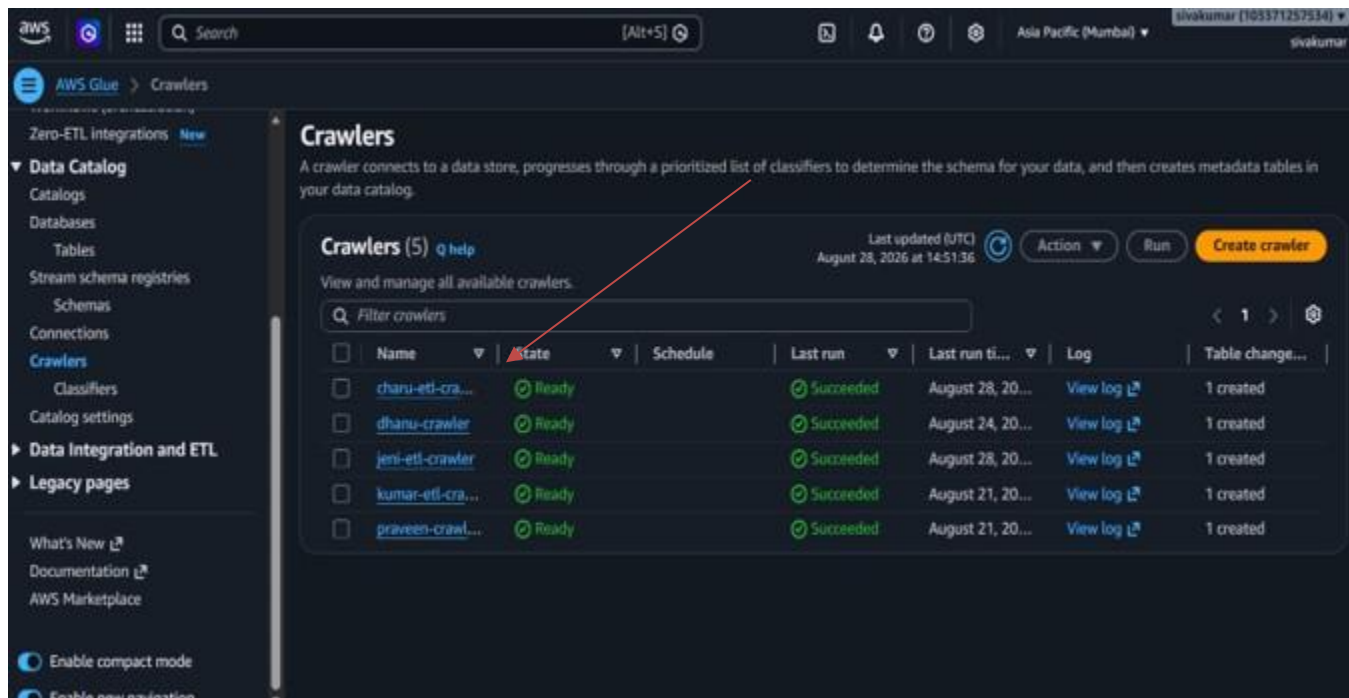
Allow the crawler to use Lake Formation credentials for crawling the data source. [Learn more](#)

☐ Use Lake Formation credentials for crawling S3 data source

Checking this box will allow the crawler to use Lake Formation credentials for crawling the data source. If the data source is registered in another account, you must provide the registered account ID. Otherwise, the crawler will crawl only those data sources associated to the account. Only applicable to S3, Glue Catalog, Iceberg, and Hudi data sources.

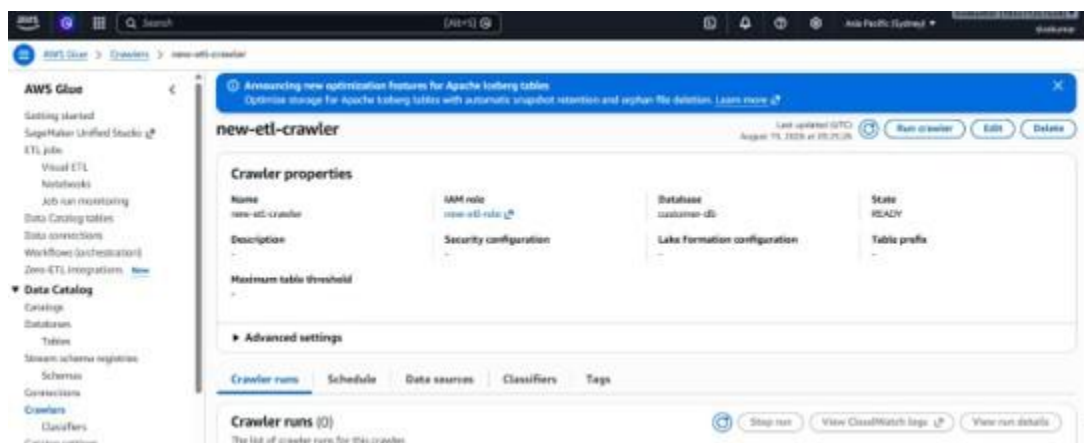
Security configuration - optional

- Set output and scheduling:**
 - Target database: selected the database created in Step 4.1
 - Table name prefix: left blank (optional)
 - Crawler schedule frequency: On demand (runs only when manually triggered)
- Reviewed all settings on the "Review and create" screen and created the crawler:**
 - Crawler name: charu-etl-crawler
 - Data source: S3 -> s3://charu-etl-source-bucket/customer-data/... (Recrawl all)
 - IAM role: new-etl-role



4.3 RUN CRAWLER AND VERIFY TABLE

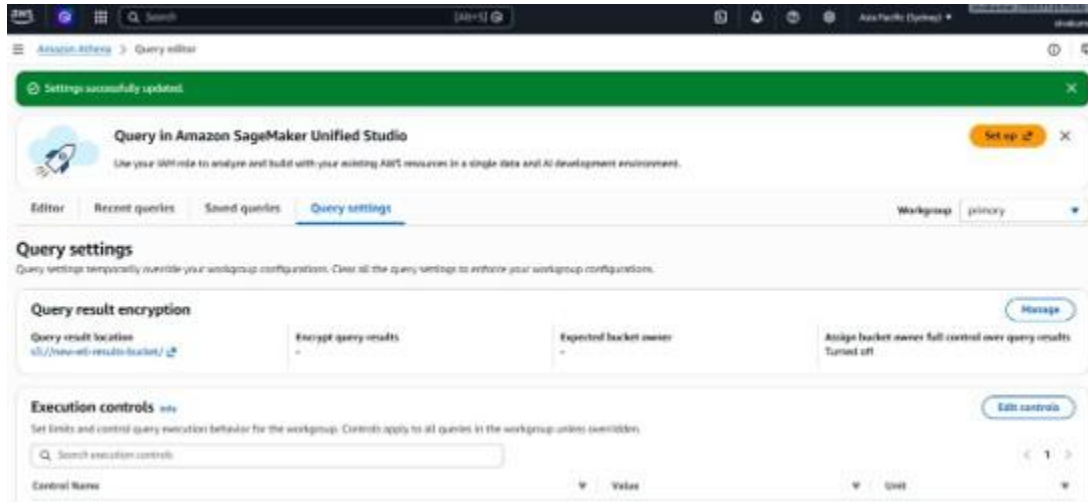
- Ran the crawler manually.
- Checked Crawler status until it showed Completed.
- Verified the table was created successfully inside the Glue Data Catalog under the selected database.



STEP 5: AMAZON ATHENA – CREATE QUERIES

- Opened Amazon Athena and set the **Query result location** to the results bucket created in Step 2 (s3://charu-etl-results-bucket /).
- Selected the database and table created by the Glue Crawler.

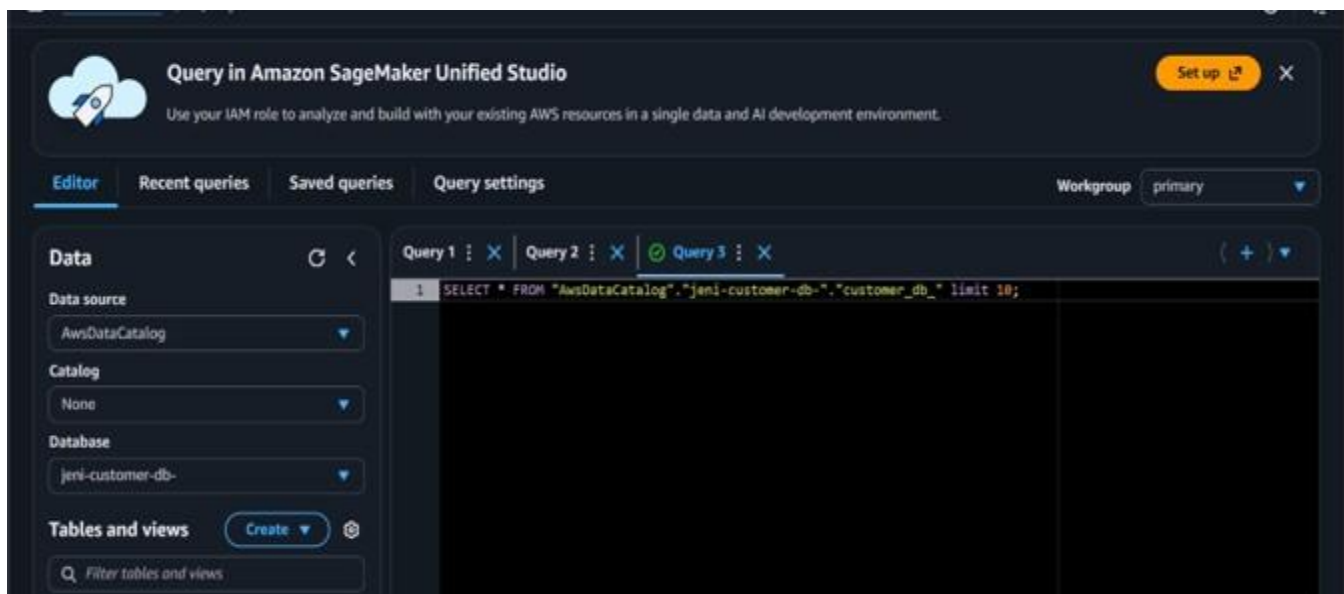
- Wrote and executed SQL queries directly on the table, including:
 - Filtering (e.g., WHERE region = 'Central')
 - Aggregate functions: MIN, MAX, AVG, SUM
 - GROUP BY analysis for region-wise and category-wise sales and profit



Amazon Athena was used as the serverless SQL layer for analysing the cataloged flight table. The database and table created by the Glue Crawler were selected before running the analytical queries.

ATHENA PREPARATION

- Opened Amazon Athena from the AWS Management Console.
- Selected the Glue Data Catalog and the database containing the flight table.
- Selected the cataloged flight table for analysis.
- Configured the S3 query-result location used by Athena.
- Executed SQL statements for booking, airline, route, fare, payment, status, city and time-based analysis.



SQL ANALYSIS GROUPS:

- **Part 1** – Basic Filtering & Aggregation
- **Part 2** – Aggregation & Grouping
- **Part 3** – Intermediate Analysis (Counts, Averages, Ranking)

AWS SQL RESULTS - PART 1 : BASIC FILTERING & AGGREGATION

1. Central Region Records

```
SELECT *  
  
FROM "AwsDataCatalog"."superstore-db"."superstore_dataset"  
  
WHERE "region" = 'Central';
```

2. First Class Shipping Records

```
SELECT *  
  
FROM "AwsDataCatalog"."superstore-db"."superstore_dataset"  
  
WHERE "ship mode" = 'First Class';
```

3. Minimum Sales in Central Region

```
SELECT MIN("sales") AS min_sales  
  
FROM "AwsDataCatalog"."superstore-db"."superstore_dataset"  
  
WHERE "region" = 'Central';
```

4. Maximum Sales in Central Region

```
SELECT MAX("sales") AS max_sales  
  
FROM "AwsDataCatalog"."superstore-db"."superstore_dataset"  
  
WHERE "region" = 'Central';
```

5. Average Sales in Central Region

```
SELECT AVG("sales") AS avg_sales  
  
FROM "AwsDataCatalog"."superstore-db"."superstore_dataset"  
  
WHERE "region" = 'Central';
```

6. Total Sales in Central Region

```
SELECT SUM("sales") AS total_sales  
  
FROM "AwsDataCatalog"."superstore-db"."superstore_dataset"  
  
WHERE "region" = 'Central';
```

AWS SQL RESULTS - PART 2 : AGGREGATION & GROUPING

7. Total Profit in Central Region

```
SELECT SUM("profit") AS total_profit
FROM "AwsDataCatalog"."superstore-db"."superstore_dataset"
WHERE "region" = 'Central';
```

8. Region-wise Total Sales

```
SELECT "region", SUM("sales") AS total_sales
FROM "AwsDataCatalog"."superstore-db"."superstore_dataset"
GROUP BY "region";
```

9. Category-wise Total Sales

```
SELECT "category", SUM("sales") AS total_sales
FROM "AwsDataCatalog"."superstore-db"."superstore_dataset"
GROUP BY "category";
```

10. Category-wise Total Profit

```
SELECT "category", SUM("profit") AS total_profit
FROM "AwsDataCatalog"."superstore-db"."superstore_dataset"
GROUP BY "category";
```

AWS SQL RESULTS - PART 3 :INTERMEDIATE ANALYSIS (COUNTS, AVERAGES, RANKING)

11. Region-wise Order Count

```
SELECT "region", COUNT(*) AS order_count
FROM "AwsDataCatalog"."superstore-db"."superstore_dataset"
GROUP BY "region";
```

12. Ship Mode-wise Order Count

```
SELECT "ship mode", COUNT(*) AS order_count
FROM "AwsDataCatalog"."superstore-db"."superstore_dataset"
GROUP BY "ship mode";
```

13. Category-wise Average Sales

```
SELECT "category", AVG("sales") AS avg_sales  
FROM "AwsDataCatalog"."superstore-db"."superstore_dataset"  
GROUP BY "category";
```

14. Category-wise Average Profit

```
SELECT "category", AVG("profit") AS avg_profit  
FROM "AwsDataCatalog"."superstore-db"."superstore_dataset"  
GROUP BY "category";
```

15. Products with Sales Greater than 500

```
SELECT *  
FROM "AwsDataCatalog"."superstore-db"."superstore_dataset"  
WHERE "sales" > 500;
```

STEP 6: SIMBA ATHENA ODBC – CONNECT ATHENA TO POWER BI

- Installed the Simba Amazon Athena ODBC driver on Windows.
- Configured a Data Source Name (DSN) with the AWS region, catalog, and workgroup details.
- This ODBC connection allows Power BI to query the Athena table directly.

STEP 7: POWER BI – CONNECT AND VISUALIZE

- Opened Power BI Desktop -> Get Data -> ODBC -> selected the configured Athena DSN.
- Authenticated and connected, then selected the Superstore table.
- Loaded the data into Power BI and used Power Query to clean it (handled nulls, mixed date formats, and data types).
- Created DAX measures for KPIs (Total Sales, Total Profit, Profit Margin, Loss Orders, etc.).
- DAX measures were created to calculate sales, profit, orders, quantity, discount, and profitability KPIs dynamically based on the selected filters.
- Additional measures such as Profit Margin %, Average Order Value, and Average Quantity per Order were used to provide deeper business insights.
- These measures were then used across the two Power BI dashboards for KPI cards, charts, maps, and tables.
- Built a two-page interactive dashboard — Executive Overview and Profitability Analysis.

PAGE 1 SALES OVERVIEW:



Part 1 — Cloud Data Infrastructure (AWS)

Service	Purpose
Amazon S3	Simple, scalable cloud storage — Source Bucket (raw data) and Result Bucket (query results)
AWS IAM	Identity Access Management — secure, role-based access control
AWS Glue	Database, Crawler -> Table — automated schema discovery and cataloging
Amazon Athena	SQL query execution — generates multiple query results for analysis

Part 2 — Data Visualization

Service	Purpose
Power BI	Connectivity via ODBC (Open Database Connectivity) — interactive data visualization